

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Question Paper Name :	IIT M DEGREE AN EXAM QDB2 15 Mar 2026
Subject Name :	2026 Mar15: IIT M AN EXAM QDB2
Creation Date :	2026-03-05 10:37:28
Duration :	120
Total Marks :	637
Display Marks:	Yes
Share Answer Key With Delivery Engine :	Yes
Actual Answer Key :	Yes
Calculator :	Scientific
Magnifying Glass Required? :	No
Ruler Required? :	No
Eraser Required? :	No
Scratch Pad Required? :	No
Rough Sketch/Notepad Required? :	No
Protractor Required? :	No
Show Watermark on Console? :	Yes
Highlighter :	No
Auto Save on Console?	Yes
Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Group I

Group Number :	1
Group Id :	64065332737
Group Maximum Duration :	0
Group Minimum Duration :	90

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	637
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

Deep Learning

Section Id :	640653129725
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653362880
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531961418

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536373441. ✓ Yes

6406536373442. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653362881

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406531961419

Correct Marks : 2

Question Label : Multiple Choice Question

Which among the following is true for a McCulloch-Pitts (MP) neuron?

Options :

6406536373443. ✗ It can implement Boolean functions with a non-linear decision boundary.

6406536373444. ✓ It can implement linearly separable Boolean functions with a linear decision boundary.

6406536373445. ✗ It can be used to approximate a real-valued function.

6406536373446. ✗ It can accurately represent the XOR function by adjusting its weights and thresholds.

Sub-Section Number :

3

Sub-Section Id :

640653362882

Question Shuffling Allowed :

Yes

Question Number : 3 Question Id : 6406531961420

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a shallow neural network with a scalar input x and a hidden layer with three neurons. The outputs of the hidden layer neurons are given by

$$h_{11} = a(b_{11} + w_{11}x),$$

$$h_{12} = a(b_{12} + w_{12}x),$$

$$h_{13} = a(b_{13} + w_{13}x),$$

where a denotes the activation function. The final output of the network is given by $y = b_{21} + w_{21}h_{11} + w_{22}h_{12} + w_{23}h_{13}$.

Suppose we multiply the parameters b_{11} and w_{11} by a constant $\alpha \neq 0$, and divide w_{21} by the same constant α , while all other parameters remain unchanged. Which of the following statements is correct?

Note:

- ReLU activation is given by $\text{ReLU}(z) = \max(0, z)$.
- Sigmoid activation is given by $\sigma(z) = \frac{1}{1 + e^{-z}}$.

Options :

6406536373447. ✓ The network output y remains unchanged if the activation function used is ReLU for $\alpha > 0, x \in \mathbb{R}$.

6406536373448. ✗ The network output y remains unchanged if the activation function used is ReLU for $\alpha < 0, x \in \mathbb{R}$.

6406536373449. ✗ The network output will be unchanged if the activation function used is sigmoid for $\alpha > 0, x \in \mathbb{R}$.

6406536373450. ✗ After the transformation, the contribution $w_{21}h_{11}$ to the output remains unchanged for all x and for all $\alpha \neq 0$.

Sub-Section Number :

4

Sub-Section Id :

640653362883

Question Shuffling Allowed :

Yes

Question Number : 4 Question Id : 6406531961421

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a multilayer perceptron (MLP) with two binary inputs x_1 and x_2 , 2 neurons in the hidden layer, and 1 in the output layer. Let h_1 and h_2 be the outputs of the hidden layer and it computes $h_1 = \text{step}(-0.5 + x_1 + x_2)$, $h_2 = \text{step}(-1.5 + x_1 + x_2)$.

The output of the network is given by $y = \text{step}(w_0 + w_1 h_1 + w_2 h_2)$, where

$$\text{step}(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}.$$

Which of the following are the possible choices of w_2 for the network to correctly compute the XOR function, given that $w_0 = -0.5, w_1 = 1$.

Options :

6406536373451. ✖ 0.5

6406536373452. ✔ -0.55

6406536373453. ✔ -0.60

6406536373454. ✖ 1

Sub-Section Number :

5

Sub-Section Id :

640653362884

Question Shuffling Allowed :

Yes

Question Number : 5 Question Id : 6406531961422

Correct Marks : 2

Question Label : Short Answer Question

Consider a neural network for a regression problem with input $x \in \mathbb{R}^{10}$. The network has ² hidden layers, each with ⁵ sigmoid neurons. The output layer has ¹ neuron and uses the sigmoid activation function. Enter the number of parameters in the network.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

91

Sub-Section Number :

6

Sub-Section Id :

640653362885

Question Shuffling Allowed :

Yes

Question Number : 6 Question Id : 6406531961423

Correct Marks : 3

Question Label : Short Answer Question

Consider a training set consisting of 12 samples. The vanilla gradient descent algorithm is used to update the model parameters over 4 epochs. The learning rate follows an exponential decay

scheme given by

$$\eta_t = 0.8 \exp(-0.2t),$$

where the iteration index t starts from zero and increments only when the parameters are

updated. What is the value of the learning rate η_t at the end of training?

(Note: Enter your answer up to 3 decimal points. For example, if your answer is 0.12345, enter 0.123).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.436 to 0.442

Sub-Section Number :

7

Sub-Section Id :

640653362886

Question Shuffling Allowed :

Yes

Question Number : 7 Question Id : 6406531961424

Correct Marks : 4

Question Label : Short Answer Question

A content recommendation system models whether a user will engage with an item based on

features $x_i \in \mathbb{R}^d$.

The training dataset is

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$$

where $y_i \in \{0, 1\}$ denotes non-engagement (0) or engagement (1). The probability of engagement is modeled using:

$$\hat{y}_i = \frac{1}{1 + e^{-(w^T x_i + b)}}$$

where $w \in \mathbb{R}^d$, and b denote the bias. The model parameters are learned using gradient descent by minimizing the squared error loss:

$$L(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Consider a training example that takes in an input vector $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, with true label $y = 1$. The

weight vector $\mathbf{w}$ is initialized to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\mathbf{b} = 0$. Compute the loss after one update of w and b for

$\eta = 1$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.08 to 0.12

Sub-Section Number :

8

Sub-Section Id :

640653362887

Question Shuffling Allowed :

No

Question Id : 6406531961425 Sub Question Shuffling Allowed : No Group Comprehension

Questions : No Question Pattern Type : NonMatrix

Question Numbers : (8 to 9)

Question Label : Comprehension

Consider the truth table of a NAND function for two binary inputs:

x_1	x_2	NAND
0	0	1
0	1	1
1	0	1
1	1	0

You use a perceptron for implementing the NAND function. The output of the neuron is given as

$$f(x) = \begin{cases} 1, & \text{if } \sum_{i=0}^2 w_i x_i \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

x_0 is assumed to be 1.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406531961426

Correct Marks : 1

Question Label : Multiple Choice Question

Is the function linearly separable?

Options :

6406536373458. ✓ Yes

6406536373459. ✖ No

Question Number : 9 Question Id : 6406531961427

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following choices for the weights w_0 , w_1 , and w_2 will produce the NAND function?

Options :

6406536373460. ✖ $w_0 = 2, w_1 = -1, w_2 = -1$

6406536373461. ✔ $w_0 = 1, w_1 = -1, w_2 = -1$

6406536373462. ✖ $w_0 = -2, w_1 = -1, w_2 = -1$

6406536373463. ✖ $w_0 = 0.5, w_1 = -0.25, w_2 = -0.25$

Sub-Section Number :

9

Sub-Section Id :

640653362888

Question Shuffling Allowed :

No

Question Id : 6406531961428 Sub Question Shuffling Allowed : No Group Comprehension

Questions : No Question Pattern Type : NonMatrix

Question Numbers : (10 to 11)

Question Label : Comprehension

Suppose you perform the perceptron algorithm on the following dataset:

x_1	x_2	y
2	1	+1
-1	-2	-1
1	3	+1
-2	-1	-1

Assume the initial weights $w_0 = [0 \ 0]^T$ and we pass the data points in the order given in the table. The following rule is used for the classification:

$$y = \begin{cases} 1, & \text{if } w^T x \geq 0 \\ -1, & \text{otherwise} \end{cases}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 10 Question Id : 6406531961429

Correct Marks : 3

Question Label : Multiple Choice Question

What will be the updated weight vector after one epoch?

Options :

6406536373464. ✖ $w = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

6406536373465. ✔ $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

6406536373466. ✖ $w = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

6406536373467. ✖ $w = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$

Question Number : 11 Question Id : 6406531961430

Correct Marks : 2

Question Label : Multiple Choice Question

After one epoch, your friend claims that the perceptron weights no longer need updating, i.e., the model has converged. Specify whether the statement is true or false.

Options :

6406536373468. ✔ True

6406536373469. ✖ False

Sub-Section Number : 10

Sub-Section Id : 640653362889

Question Shuffling Allowed : No

Question Id : 6406531961431 Sub Question Shuffling Allowed : No Group Comprehension

Questions : No Question Pattern Type : NonMatrix

Question Numbers : (12 to 13)

Question Label : Comprehension

Consider a neural network for a binary classification problem with one input and one output. There is one hidden layer with two neurons and an output layer. Both the layers uses sigmoid as an activation function. Cross entropy loss is used.

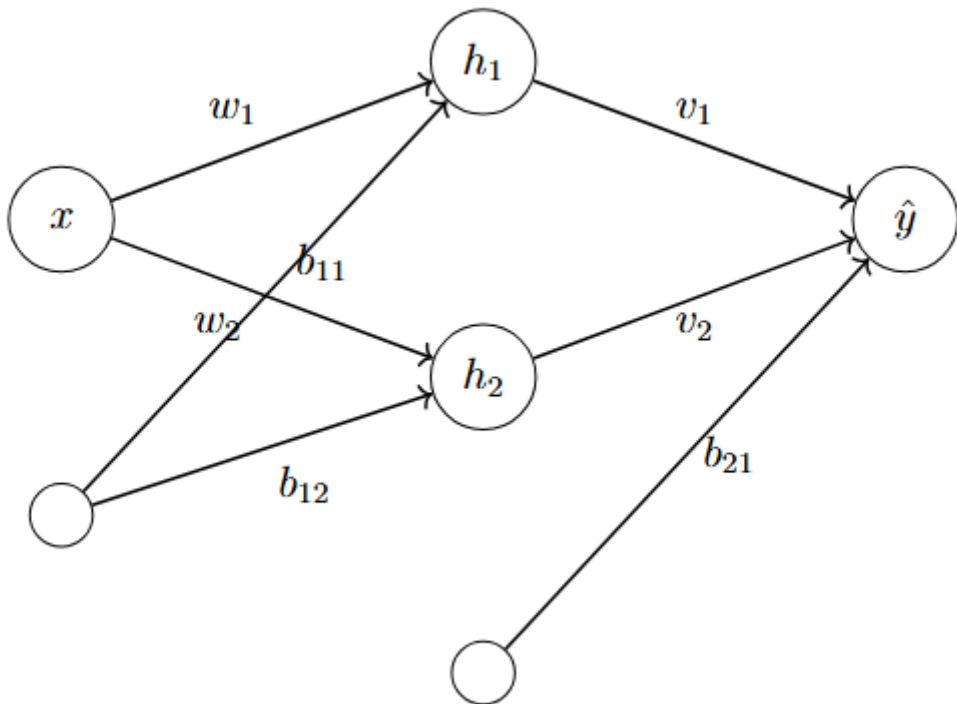
Following are the values of weights and biases:

$$w_1 = 1, \quad b_{11} = 0,$$

$$w_2 = -1, \quad b_{12} = 0,$$

$$v_1 = 1, \quad v_2 = 1,$$

$$b_{21} = 1, \quad x = 1, y = 1.$$



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 12 Question Id : 6406531961432

Correct Marks : 3

Question Label : Short Answer Question

Compute the predicted output $\hat{y}$ for the given input x . Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.86 to 0.90

Question Number : 13 Question Id : 6406531961433

Correct Marks : 4

Question Label : Short Answer Question

Compute

$$\frac{\partial L}{\partial w_1}.$$

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-0.04 to 0.00

Sub-Section Number : 11

Sub-Section Id : 640653362890

Question Shuffling Allowed : No

Question Id : 6406531961434 Sub Question Shuffling Allowed : No Group Comprehension

Questions : No Question Pattern Type : NonMatrix

Question Numbers : (14 to 15)

Question Label : Comprehension

Consider a linear regression model $\hat{y} = wx$ trained using momentum-based mini-batch gradient descent with the loss function:

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (wx_i - y_i)^2,$$

where m is the batch size. The algorithm uses a mini-batch size of 2 , learning rate $\eta = 0.1$, and the momentum coefficient $\beta = 0.9$. The weights and velocity are initialized as $w_0 = 1$ and $u_{-1} = 0$. The update rules are given by:

$$u_t = \beta u_{(t-1)} + \nabla J(w_t)$$

$$w_{(t+1)} = w_t - \eta u_t.$$

The first mini-batch consists of the following samples:

x	y
1	2
2	4

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 14 Question Id : 6406531961435

Correct Marks : 3

Question Label : Multiple Choice Question

Compute the gradient $\nabla J(w_0)$ for this mini-batch.

Options :

6406536373472. ✓ -2.5

6406536373473. ✗ -5.0

6406536373474. ✗ -7.5

6406536373475. ✗ 5.0

Question Number : 15 Question Id : 6406531961436

Correct Marks : 3

Question Label : Short Answer Question

Using the gradient computed in the previous question, determine the velocity u_0 after the first iteration.

Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-2.5